

Modeling crystallization kinetics for selective laser sintering of polyamide 12

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Abstract

Selective laser sintering (SLS) of polymers represents a widely used additive manufacturing process, where the part quality depends highly on the present thermal conditions. One distinct feature of SLS is the existence of separate temperature regions for melting and crystallization (solidification) and that the process optimally operates within said regions. Typically a crystallization model, such as the Nakamura model, is used to predict the degree of crystallization as a function of temperature and time. One limitation of this model is the inability to compute negative rates of the crystallization degree during remelting. As we will show in this work, such an extension is necessary, considering the varying temperature fields appearing in SLS. To this end, an extension is proposed and analyzed in detail. Furthermore, a dependency of the temperature and crystallization fields on the size of geometrical features is presented.

KEYWORDS

additive manufacturing, crystallization kinetics, numerical simulation, selective laser sintering

1 | INTRODUCTION

Simulation for additive manufacturing (AM) of polymers represents a field of high research activity covering several approaches, such as extrusion methods [4,11,29] or selective laser sintering (SLS) of polymers [1,3,10,22–25,28]. For an overview of additive manufacturing using polymers we refer to [5–7,19,27] and references therein. In SLS of polymers, three-dimensional parts are built in a layer wise fashion. The classic process model consists of the following stages: In a preheating stage, the building chamber, that keeps the powder material, is heated to a specified build temperature by means of a heating system, for example, radiators. Thereafter, in the build stage, powder material is selectively melted via the energy provided by a laser, followed by the lowering of the build platform and the deposition of a new powder layer. These steps are carried out in a repetitive manner until every cross-section of the part has been melted, while being supported by the surrounding powder. It is further assumed that during this phase, the molten material rests at the specified build temperature (isothermal condition). In the third stage, that is, cooling stage, the building chamber is cooled down and crystallization of the molten material is induced [7].

As pointed out by [7,26] the validity of this isothermal model is very limited. Indeed, besides a heat accumulation in the molten region, the influence of powder deposition and successive melting, that lead to temperature variations, can be detected even multiple layers below the surface, rendering a nonisothermal process. Hence, a new process model was proposed in [7], which accounts for the temperature gradient in build direction. Furthermore, premature prediction of full

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crystallization and a size dependency was observed in [26]. These observations additionally contribute to the remaining challenges in SLS, for instance warpage and residual stresses, reproducibility and quality of mechanical parameters [6]. These stem from complex thermal conditions along with the sensitivity of the process with respect to temperature. Therefore, simulation tools are essential to deepen the process understanding and to investigate compensation approaches to reduce these undesired effects.

In [26] a verified and validated thermal solver was presented that includes the modeling of crystallization kinetics. The crystallization model was adapted to allow for negative crystallization rates to account for remelting of partially crystallized material. This becomes necessary since the deposition of fresh powder, which is typically at a lower temperature than the build temperature, lowers the temperature, followed by a remelting due to the laser beam.

In this contribution, the crystallization model and its numerical implementation are thoroughly analyzed. Different approaches are investigated and the applicability of the model within the SLS process simulation is demonstrated by means of numerical examples.

The structure of this work is as follows. In Section 2 we briefly discuss the governing equations used to solve for the temperature fields and the degree of crystallization, including the material model for the herein considered material Polyamide 12 (PA12). In Section 3 the discretization in space and time is discussed, followed by numerical examples in Section 4 and some concluding remarks in Section 5.

2 | MODELING SELECTIVE LASER SINTERING

This section summarizes the continuum model used to compute the temperature and crystallization fields, as well as the material parameters.

2.1 | Balance of energy

To compute the evolution of the temperature field ϑ in a continuum body $\mathcal{B}$, the heat equation in the form given in Equation (1) is solved

$$\dot{h}(\vartheta) \rho = \nabla \cdot (\lambda(\vartheta) \nabla \vartheta) + Q + \dot{\alpha} \rho \Delta h_{\alpha}. \quad (1)$$

It corresponds to a mixed temperature-enthalpy formulation based on the method presented in [15]. The specific enthalpy is given by $h(\vartheta)$, which includes the effect of latent heat of fusion described by the material parameter Δh_m (see Figure 4). Furthermore, the thermal conductivity and the mass density are denoted by $\lambda(\vartheta)$ and ρ respectively. The power input due to the laser beam is denoted by the term Q . The last term in Equation (1) accounts for the exo-thermal reaction during crystallization. It consists of the rate of the relative degree of crystallization $\dot{\alpha}$, with $\alpha \in \{0, 1\}$, the mass density and the specific energy released during crystallization Δh_{α} .

The balance equation is complemented by boundary conditions of Dirichlet and Neumann type. Regarding the Neumann type boundary conditions on the corresponding boundary $\partial \mathcal{B}^N$, with $q = \lambda \nabla \vartheta \cdot \mathbf{n} = q_1 + q_2$, we consider convection

$$q_1 = \mathcal{H} [\vartheta^{\text{conv}} - \vartheta], \quad (2)$$

where $\mathcal{H}$ denotes the heat transfer coefficient and ϑ^{conv} a reference temperature, and radiation with

$$q_2 = \sigma \epsilon [\vartheta^{\text{rad}4} - \vartheta^4], \quad (3)$$

where σ corresponds to the Stefan–Boltzmann constant, ϵ to the emission coefficient and ϑ^{rad} to a PID controlled reference temperature to mimic the effect of the heating system in the building chamber (see [26] for details).

The effect of the energy input due to a laser beam is accounted for by a heat source term Q , which corresponds to a Gaussian power distribution based on [23]:

$$Q(x, y, z, t) = \frac{2 P_b \eta}{I r_b^3 \sqrt{2\pi}} \exp \left(-2 \left[\frac{[x - x_0(t)]^2}{r_b^2} + \frac{[y - y_0(t)]^2}{r_b^2} + \frac{[z - z_0(t)]^2}{r_b^2} \right] \right) \quad (4)$$

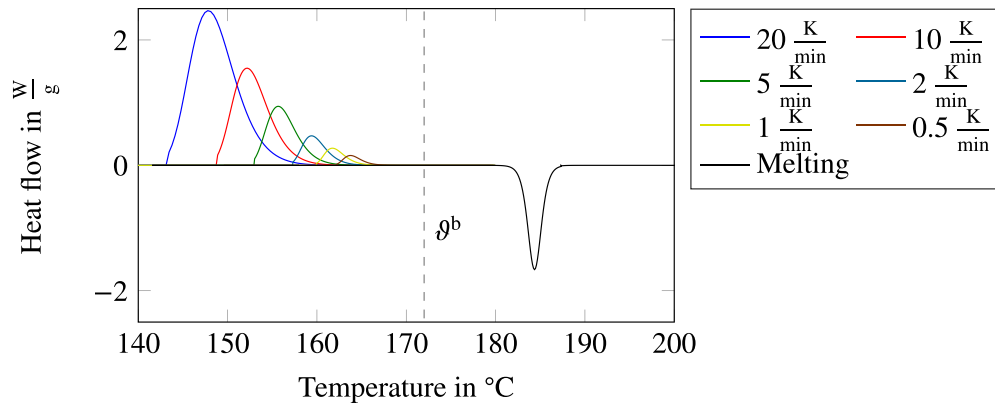


FIGURE 1 The heat flow during heating and cooling, indicating the temperature regions in which melting and crystallization take place. The desired build temperature during SLS (e.g., $\vartheta^b = 172^\circ$) lies between these two temperature regions. Furthermore, the presence of multiple positive heat flow curves indicates the dependence of crystallization on temperature and time. The values for the crystallization were taken from [26]. The initial temperature was set to $\vartheta_0 = 180^\circ$ followed by a cooling with various rates as given in the legend

Here, P_b and η denote the power and absorption coefficient respectively. The integral I ensures energetic consistency. The beam radius is indicated by r_b . The beam center $(x_0(t), y_0(t), z_0(t))$ moves along the scan path with the velocity v_b .

The radius of the laser beam r_b is typically in the order of μm , for example, $200 \mu\text{m}$, as is the so-called hatching distance. This is the distance between distinct lines that the laser beam moves while melting the current cross-section. For a relatively small part geometry of $20 \times 20 \times 20 \text{ mm}^3$ and a layer height of $100 \mu\text{m}$ this renders a total length of the scan path of approximately 400 m. Under the assumption that the center of the heat source term Q is moved by $100 \mu\text{m}$ in every time step, this would render a total of 4×10^6 time steps, required to resolve the scan path. To reduce the associated numerical cost we follow [12], and average the effect of the laser beam over time.

$$\bar{Q}(x, y, z, t) = \frac{1}{\Delta t} \left[\int_{t_0}^{t_1} Q dt + \int_{t_1}^{t_2} Q dt + \dots + \int_{t_{n-1}}^{t_0 + \Delta t} Q dt \right]. \quad (5)$$

This leads to a series of heat input lines, for example, from t_0 to t_1 or from t_{n-1} to $t + \Delta t$, that are applied concurrently. This allows for a simple parametrization of the total length of the sum of heat input lines by the time step size Δt to choose between a moving point source, melting parts of the cross-section at once, or melting the whole cross-section within one time step. As a measure of the total relative length of these heat input lines [12] proposed

$$\tau = \frac{v_b \Delta t}{r_b}, \quad (6)$$

which will be used in Section 4. The suitability of this approach, especially for the low conducting polymers considered here, has been shown in [25,26].

2.2 | Crystallization model

The modeling of crystallization kinetics has been the objective of investigation for quite some time and several approaches for different materials are available in literature [1,17,18,21,30]. The model used herein and the proposed adaption will be discussed in the following.

During the SLS process powder material is selectively molten by means of a laser beam. The transition from the molten state to a solidified state does not follow the same path as the transition from solid (powder) to the molten state. This can be described by the schematic graph shown in Figure 1 in which the heat flow during melting and crystallization is shown. The positive and negative heat flows are due to the crystallization and melt enthalpy, that is, Δh_α and Δh_m , and indicate the temperature ranges in which crystallization or melting, respectively, occur.

The desired build temperature ϑ^b lies within the region between the temperatures where melting occurs and the onset temperature of crystallization. This leads to a co-existence of molten and powder material during the SLS build stage [30].

While the melting is assumed to be purely temperature dependent, the crystallization depends on temperature and time. This becomes apparent considering the different heat flows for different thermal conditions. These curves are usually determined by experiments using differential scanning calorimetry (DSC) and we refer for instance to [1,30] for a more detailed discussion on the experimental procedure. It should be noted that crystallization can also be induced at a constant temperature, that is, a cooling rate of 0 K/min, and the crystallization behavior depends on the set temperature. Furthermore, the semi-crystalline polymer PA12 considered within this work is characterized by the existence of an amorphous next to a crystalline phase in the solidified state and we only consider the relative degree of crystallization throughout this work, that is, values between 0 and 1.

As mentioned in Section 2.1, the exo-thermal crystallization of molten polymer powder contributes in Equation (1) via $\dot{\alpha}\rho\Delta h_\alpha$. This term consists of an energetic quantity Δh_α that represents the total specific energy that is being released during crystallization, the density ρ and the rate of the relative degree of crystallization $\dot{\alpha}$ (for the remaining material parameters see Section 2.3). In this work the quantity Δh_α is assumed to be constant which is related to a constant absolute degree of crystallization [1,16].

The relative degree of crystallization is computed based on the Nakamura model [20] as presented in [30]

$$\alpha = 1 - \exp\left(\left[\int_0^t K(\vartheta) d\tilde{t}\right]^n\right) \quad (7)$$

with n denoting the Avrami exponent and K the so-called modified crystallization rate. To compute α , [21] pointed out that a rate formulation as given in Equation (8) is frequently chosen, that is, using

$$\dot{\alpha} = K(\vartheta) G(\alpha) \geq 0 \quad (8)$$

where the terms $G(\alpha)$ and $K(\vartheta)$ read

$$G(\alpha) = n[1 - \alpha][-\ln(1 - \alpha)]^{1-\frac{1}{n}} \quad (9)$$

$$K(\vartheta) = \ln(2)^{\frac{1}{n}} K_0 \exp\left(-\frac{U}{R[\vartheta - \vartheta_\infty]}\right) \exp\left(\frac{K_g[\vartheta + \vartheta_m^0]}{2\vartheta^2[\vartheta_m^0 - \vartheta]}\right). \quad (10)$$

Thereby K_0 denotes a so-called growth rate constant, U the activation energy for polymer diffusion, R the universal gas constant, K_g the nucleation rate constant, $\vartheta_\infty = \vartheta_g - 30$ K, with ϑ_g denoting the glass transition temperature, and ϑ_m^0 the equilibrium melting temperature (see [1,30] for details). The corresponding material parameters are summarized in Section 2.3.

Furthermore, a so-called nucleation value is necessary, since Equation (8) would never be larger than zero with $\alpha = 0$. This is also apparent considering Figure 2. A value of $\alpha = 0$ would lead to a value of $G(0) = 0$ which in turn would lead to a value of $\dot{\alpha} = 0$. Since G depends on α , and α grows based on Equation (8), α would rest at a value of 0 regardless of the thermal conditions.

As has been shown in [17], $\alpha_{\min} = 1e - 6$ and below are suitable nucleation values. If $\alpha_{\min}$ increases, the prediction of the time of full crystallization decreases since it acts as a nucleation value. The value $\alpha_{\max}$ is merely used for the tangent approximation in the vicinity of $\alpha = 1$ since the derivative with respect to α tends towards infinity at $\alpha = 1$.

In this work these values are set to $\alpha_{\min} = 1e - 6$ and $\alpha_{\max} = 1 - \alpha_{\min}$.

In SLS the building chamber is heated to a desired process temperature, for example, $\vartheta^b = 172^\circ$ where the material then undergoes frequent temperature changes. The temperature rises due to the energy input of the laser beam which melts the current layer. Since the fresh powder is kept at a temperature that is below the process temperature (to increase re-usability of the powder), the temperature in the process decreases upon the deposition of a new powder layer.

The ansatz as given in Equation (8) prohibits a possible decrease of the crystallization degree α due to the inability of Equation (8) to compute negative rates. This would be necessary in the case of remelting of partially crystallized material, which can be observed in the previously molten layers that are still affected by the energy input of the laser to the current layer, as shown in [7,26].

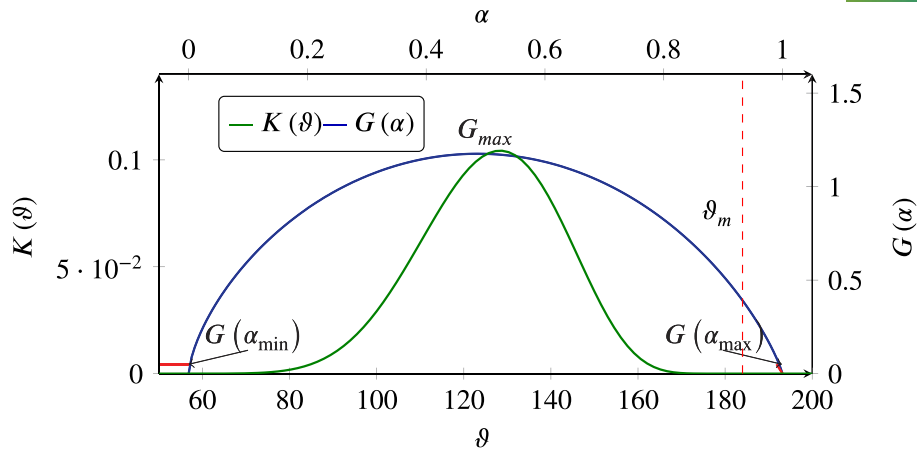


FIGURE 2 Illustration of the functions K and G . The value $\alpha_{\min}$ serves as a nucleation value to prohibit a constant zero rate of crystallization if $\alpha = 0$ (since $G(0) = 0$). The value $\alpha_{\max}$ is used for a tangent approximation to improve numerical robustness (see [17] for details.). The value ϑ_m refers to the melting temperature

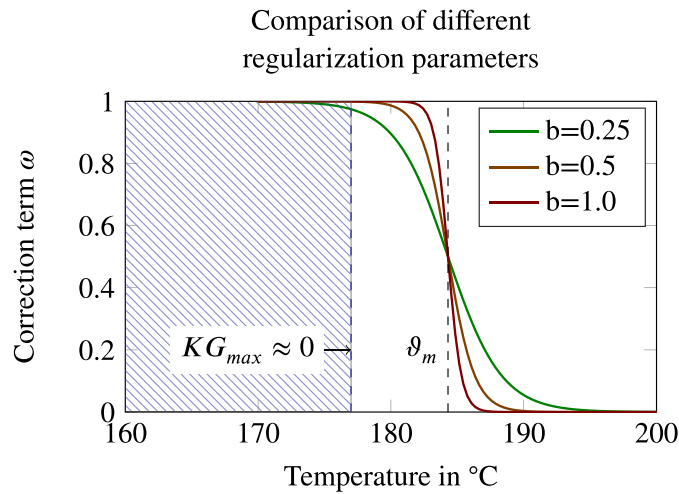


FIGURE 3 Illustration of the regions of the modified model (Eq. (11)). For a sufficiently large regularization parameter b , the correction ω does not interfere with the original rate formulation from Equation (8) and vice versa. For instance, at a temperature of $\vartheta = 177^\circ$ the rate of Equation (8) is effectively zero even for the case that G reached its maximum value (see Figure 2). The value ϑ_m denotes the melting temperature

To circumvent this behavior and allow for negative rates of α we introduce a modification of equation (8) as

$$\dot{\alpha} = K(\vartheta)G(\alpha) + \alpha^* \frac{d\omega}{d\vartheta} \dot{\vartheta}, \quad \text{with } \omega = \begin{cases} 1, & \dot{\vartheta} \leq 0 \\ \left[\frac{1}{2} [\tanh(b[\vartheta_m - \vartheta]) + 1] \right], & \dot{\vartheta} > 0 \end{cases}, \quad (11)$$

which preserves the original crystallization model for $\dot{\vartheta} \leq 0$ and introduces a modified rate for $\dot{\vartheta} > 0$. This modification reduces the degree of crystallization α if the temperature reaches the melting temperature. The correction term ω is illustrated in Figure 3, for different values of the parameter b . Since $K \rightarrow 0$ if $\vartheta \rightarrow \vartheta_m$, this modification does not interfere with the original crystallization model as long as the regularization parameter b in the correction term ω is sufficiently large. This can be seen since the two regions are otherwise clearly separated from each other.

The basic idea is to account for the temperature dependent behavior during melting via guiding α using ω . The term α^* corresponds to the relative degree of crystallization solely using the rate for the case of $\dot{\vartheta} \leq 0$, that is, the unmodified

TABLE 1 Summary of the used parameters for the crystallization model and melting based on [1,30]

U	6279 J/mol	R	8.314 J/(mol K)
K_g	136235 K ²	K_0	37869 1/s
ϑ_m^0	193.2°	n	3
Δh_m	9.1×10^4 J/kg	Δh_α	4.7×10^4 J/kg

value as indicated in equation (11). Hence, it represents an additional history variable. For instance, if the material would crystallize to a value of $\alpha = 0.5$ and the temperature would rise to the melting temperature ϑ_m , the correction would reduce the degree of crystallization by 50 % since $\omega(\vartheta_m) = 0.5$. Yet, the term obtained using Equation (8), that is, α^* , has not been altered since it could only increase during crystallization. However, if a sequence of (partial) crystallization and melting processes are considered, it could be required to reduce the value of α^* . This is necessary if the temperature leaves the melting region, that is, the temperature region in which $0 < \omega < 1$, and is captured by the following update

$$\alpha^* = \begin{cases} \alpha^*, & \frac{d\omega}{d\vartheta} < 0 \\ \alpha, & \frac{d\omega}{d\vartheta} = 0 \end{cases} \quad (12)$$

This means that the unmodified value is kept constant in the melting region or updated to the current value of α if either the material is fully molten or the temperature is at a region where the unmodified crystallization model is employed.

It should be noted that thermodynamic consistency for both, the unmodified and the modified crystallization model, can be shown using a specific free energy for the crystallization part as, for instance, proposed in [9,14], that is, $\Psi_\alpha = \Delta h_\alpha \frac{\vartheta - \vartheta_{on}}{\vartheta_{on}} \alpha$. The reduced dissipation inequality (RDI) reads $-\frac{\partial \Psi_\alpha}{\partial \alpha} \dot{\alpha} \geq 0$ with $\frac{\partial \Psi_\alpha}{\partial \alpha} = \Delta h_\alpha \frac{\vartheta - \vartheta_{on}}{\vartheta_{on}}$. The limit for the rate dependent onset temperature ϑ_{on} for crystallization is indicated in Figure 3 as the boundary of the blue area. During crystallization the temperature is below the onset temperature, rendering, with $\dot{\alpha} \geq 0$, the RDI satisfied. During remelting the temperature is above ϑ_{on} while the rate of crystallization $\dot{\alpha} \leq 0$, rendering the RDI also satisfied (given $\frac{d\omega}{d\vartheta} = 0$ for $\vartheta \leq \vartheta_{on}$).

2.3 | Material parameters

For the numerical discretization, which will be discussed in Section 3, we employ the finite element method (FEM) and allow for different states of every element, that is, powder, solid or melt. The transition from a powder element to a melt element requires that the average temperature of the corresponding quadrature points lies above the melting temperature. The transition from melt to solid requires that the average of the degree of crystallization at the quadrature points is greater or equal $\alpha_{\max} - 1 \times 10^{-6}$. The corresponding modeled material parameters are presented here for these three states.

The values for specific enthalpy and thermal conductivity for the respective states are presented in Figure 4. Throughout this work we keep the density ρ constant at a value of $\rho = 1020$ kg/m³. An element can have the powder status only once (until it switches to melt), while it could potentially change between solid and melt several times without restrictions. The parameters for the crystallization model and the latent heat of fusion are based on [1,30] and summarized in Table 1.

3 | NUMERICAL ASPECTS

In this section the discretization in space and time are shortly discussed. Therefore Equation (1) is transformed to the weak form which reads

$$\mathcal{R} = \int_B [\delta \vartheta \dot{h} + \lambda \nabla \delta \vartheta \cdot \nabla \vartheta - \delta \vartheta Q - \delta \vartheta \dot{\alpha} \rho \Delta h_\alpha] dV - \int_{\partial B^N} \delta \vartheta [q_1 + q_2] dA \stackrel{!}{=} 0 \quad \forall \delta \vartheta \in \mathcal{V} \quad (13)$$

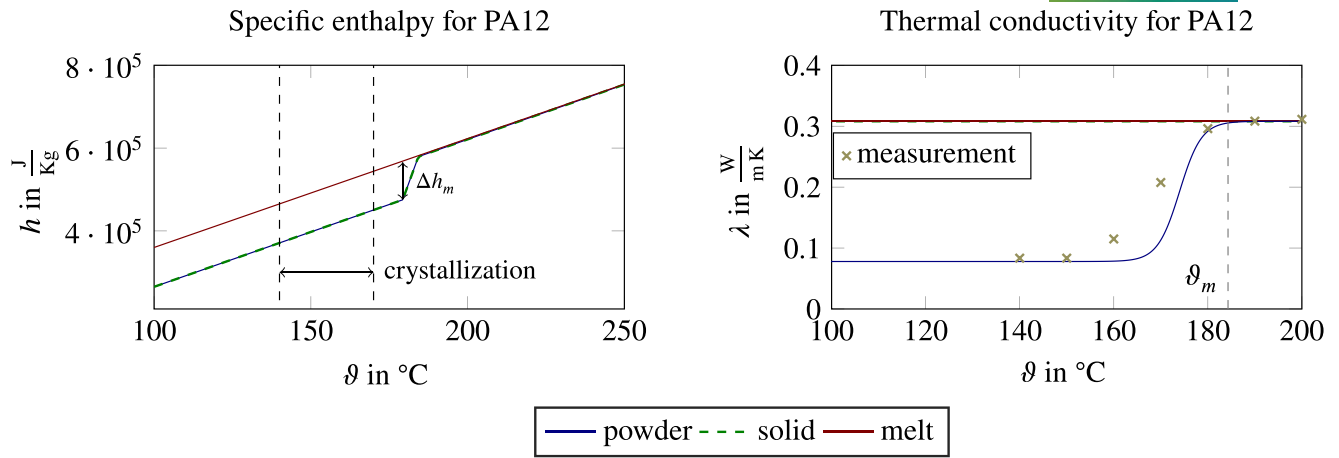


FIGURE 4 Material parameters for PA12. Depending on the state of every element the corresponding curve is selected. The decision whether or not an element changes from melt to solid depends on the relative degree of crystallization α . Once an element switched from powder status to melt, it can only change further from melt to solid and back to melt. The measurements were taken from [23]. The melting temperature was set to $\vartheta_m = 184.3^\circ$

with $\delta\vartheta$ denoting the test function, $\mathcal{V}$ the Sobolev space of test functions and $\partial\mathcal{B}^N$ the boundary subject to Neumann type boundary conditions. It should be emphasized that the primary unknown is the temperature field ϑ , while both, the specific enthalpy h and the degree of crystallization α , live on material point level, while depending on the temperature.

3.1 | Discretization in time

For the temporal discretization the time derivatives in Equation (13), that is, of the specific enthalpy and the relative degree of crystallization, need to be discretized. A frequently used first order discretization scheme is the backward Euler method. This represents an implicit method and requires the solution of

$$\mathcal{R} \left(\frac{h_n(\vartheta_n) - h_{n-1}(\vartheta_{n-1})}{\Delta t}, \vartheta_n, \frac{\alpha_n(\vartheta_n) - \alpha_{n-1}(\vartheta_{n-1})}{\Delta t} \right) \stackrel{!}{=} 0 \quad (14)$$

for an evolution from time t_{n-1} to t_n using the time step size Δt . Recall that the primary unknown is the temperature field ϑ and the residual is solved for ϑ , while h and α merely live on point level and depend on the temperature.

The temporal discretization used herein is a 2 stage singly diagonal Runge–Kutta (S-DIRK) scheme as introduced in [8]. Besides a higher convergence order, this approach provides a temporal error estimation which allows for an adaptive time step size control. Since there is a unique relation between ϑ , the primary unknown, and the specific enthalpy h , and the rate of h appears in the balance equation, we use $\dot{h}$ for the time step size control. This is crucial to bridge the scales between the time scale during melting ($\sim$ ms) and the remaining process stages ($\sim$ min or h). Instead of directly solving from time t_{n-1} to t_n , the RK scheme introduces the notion of so-called stages i . These represent time instances within the time interval t_{n-1} to t_n . Application of the 2 stage S-DIRK method to Equation (13), requires the solution of the two residual equations

$$\mathcal{R} \left(\frac{h_i(\vartheta_i) - h_{s_i}}{a_{ii}\Delta t}, \vartheta_i, \frac{\alpha_i - \alpha_{s_i}}{a_{ii}\Delta t} \right) \stackrel{!}{=} 0 \text{ for } i = 1, 2 \quad (15)$$

where a_{ij} are values from a corresponding Butcher tableau. For the S-DIRK method, the corresponding Butcher tableau has a lower triangular structure, allowing for a decoupled solution of Equation (15) for the respective stages. The equilibrium is computed at both stages, that is, $i = 1$ and $i = 2$, where the solution of the second stage coincides with the solution of the whole time step, for example, $\vartheta_{i=2} = \vartheta_n$. To compute the approximations of the time derivatives so-called

start values $(\bullet)_{s_i}$ are required:

$$h_{s_1} = h_{n-1} \quad \text{and} \quad h_{s_2} = h_{n-1} + \frac{a_{21}}{a_{11}} [h_1 - h_{s_1}] \quad (16)$$

$$\alpha_{s_1} = \alpha_{n-1} \quad \text{and} \quad \alpha_{s_2} = \alpha_{n-1} + \frac{a_{21}}{a_{11}} [\alpha_1 - \alpha_{s_1}] \quad (17)$$

Basically one uses the rate computed in the first stage to compute a start value for the second stage. The solution of the second stage then coincides with the desired value at time step n (see [8] for details).

3.2 | Discretization in space

In order to compute the temperature fields, we discretize Equation (15) in space using the FEM with bi-linear hexahedral elements (based on the ansatz proposed in [15]). The primary unknown is the temperature field while the time derivatives of both, the specific enthalpy as well as the degree of crystallization are computed on quadrature point level.

The implementation is carried out in an in-house C++ code which makes use of the open source library *deal.ii* [2]. For a multitude of verification and validation examples, as well as a deeper discussion about the employed spatial adaptivity methods, we refer to [26]. After discretization of Equation (15) this eventually leads to the requirement to solve at every RK stage i a nonlinear vector-valued equilibrium equation of the form

$$\mathbf{R}_i = \mathbf{0}, \quad \text{for } i = 1, 2 \quad (18)$$

which is solved by means of the Newton–Raphson (NR) method. In addition to this global nonlinear equation, the degree of crystallization requires an additional (local) iterative solution procedure. This is due to the dependency of the crystallization rate on both, ϑ and α (see Eqs. (9) and (10)), which will be discussed in the sequel.

3.3 | Remarks on the time discretization of α

As pointed out in Section 3.2 the degree of crystallization α is treated as an internal variable, living on quadrature point level. For the 2 stage RK scheme the following nonlinear equation has to be solved (e.g., using NR) on every quadrature point for every stage i :

$$\alpha_i = \alpha_{s_i} + a_{ii} \Delta t \dot{\alpha}_i \quad \text{for } i = 1, 2 \quad (19)$$

with

$$\dot{\alpha}_i = \begin{cases} K(\vartheta_i) G(\alpha_i), & \frac{\vartheta_i - \vartheta_{s_i}}{\Delta t} \leq 0 \\ K(\vartheta_i) G(\alpha_i) \omega(\vartheta_i) + \alpha_i^* \frac{d\omega(\vartheta_i)}{d\vartheta} \frac{\vartheta_i - \vartheta_{s_i}}{\Delta t}, & \frac{\vartheta_i - \vartheta_{s_i}}{\Delta t} > 0 \end{cases} \quad (20)$$

and the update of α^*

$$\alpha_i^* = \begin{cases} \alpha_{s_i}^*, & \frac{d\omega(\vartheta_{s_i})}{d\vartheta} < 0 \\ \alpha_{s_i}, & \frac{d\omega(\vartheta_{s_i})}{d\vartheta} = 0 \end{cases} \quad (21)$$

As will be shown in Section 4.1 this formulation requires a sufficiently small time step size in order to avoid jumps over the temperature range in which the correction term ω is active. To circumvent this limitation, the formulation is slightly modified, rendering a sort of integral formulation:

$$\dot{\alpha}_i = \begin{cases} K(\vartheta_i) G(\alpha_i), & \frac{\vartheta_i - \vartheta_{s_i}}{\Delta t} \leq 0 \\ K(\vartheta_i) G(\alpha_i) \omega(\vartheta_i) + \frac{\omega(\vartheta_i) \alpha_i^* - \alpha_{s_i}}{\Delta t}, & \frac{\vartheta_i - \vartheta_{s_i}}{\Delta t} > 0 \end{cases} \quad (22)$$

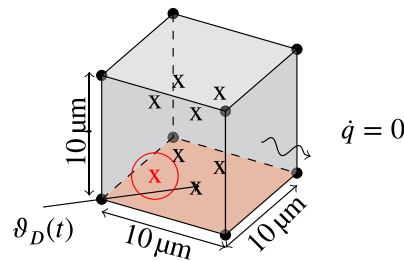


FIGURE 5 Setup for the one-element-tests to study the crystallization model. The values are recorded at the indicated (red) quadrature point of the element

With this modification jumps over said temperature range still lead to a reduction of the relative degree of crystallization α . In case of high temperature rates or large time step sizes, convergence issues can occur solving for the current degree of crystallization. To decouple the requirements on the time step size of the temperature field and the crystallization field, we allow for sub-time stepping. More specifically, if convergence issues occur in either of the RK stages while solving the local NR scheme for degree of crystallization, the time step size at quadrature point level is divided by a certain number, for example, 10, and the corresponding values of the temperature are interpolated linearly. If necessary this procedure can be repeated. The choice of the number to divide by is rather heuristic but proved sufficient in the present work.

4 | EXAMPLES

In the sequel numerical examples for the different approaches outlined in this work are presented. First, investigations on quadrature point level are carried out by means of a one-element-test, followed by a quasi one-dimensional problem. Thereafter, the temperature and degree of crystallization evolution will be computed for a larger geometry in Section 4.3.

4.1 | One-element-test

In this section, the one-element-test is briefly summarized. In this case a very small element is used in the order of the size of one powder particle. The size of the element dictates the heat conduction, that is, the larger the element the larger the potential thermal gradient. To be able to have control over the temperature without prescribing it on quadrature point level, a very small element is employed. This allows to simultaneously investigate the different discussed approaches and the implementation, that is, rather than verifying only the material model by prescribing the temperature directly at quadrature point level, the used approach allows to simultaneously verify and validate the finite element implementation at least with regard to crystallization, by simply changing the boundary conditions.

The setup is outlined in Figure 5. Accordingly, a prescribed (possibly transient) Dirichlet-type boundary condition ϑ_D is applied at the indicated bottom face. The remaining faces are subject to homogeneous Neumann-type boundary conditions. The quantities that will be investigated in the following, for example, the degree of crystallization, are measured at the quadrature point highlighted in red.

4.1.1 | ω correction—Influence on α Evolution

The first example considers the effect of the correction term ω on the evolution of α and the energetic consistency of the model. Figure 6 presents the temperature value and the corresponding accumulated released heat for the one-element-test, comparing the modified and unmodified model, that is, Equation (8) versus Equation (11) (the unmodified model is indicated by the constant value $\omega = 1$). Although at some point the temperature exceeds the melting temperature, the unmodified model of Equation (8) leads to a stagnation of the degree of crystallization and the corresponding accumulated released heat.

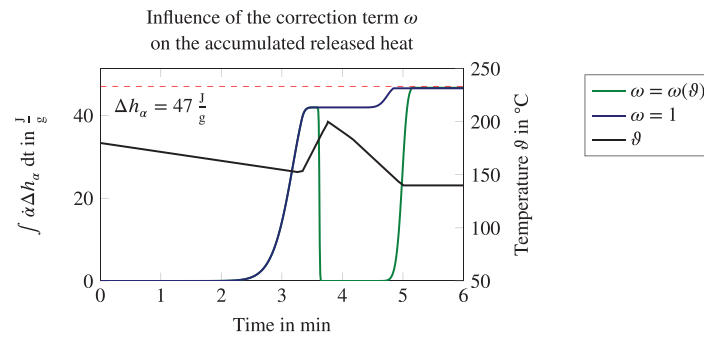


FIGURE 6 Comparison of the model corresponding to Equation (8) ($\omega = 1$) and the modified model from Equation (11) ($\omega = \omega(\theta)$) in terms of the accumulated released heat for a given temperature field. At a fully crystallized state, the accumulated released heat matches the total energy released during the exo-thermal reaction, that is, Δh_α , indicated by the red line. The time step size for the computation was set to $\Delta t = 0.5$ s

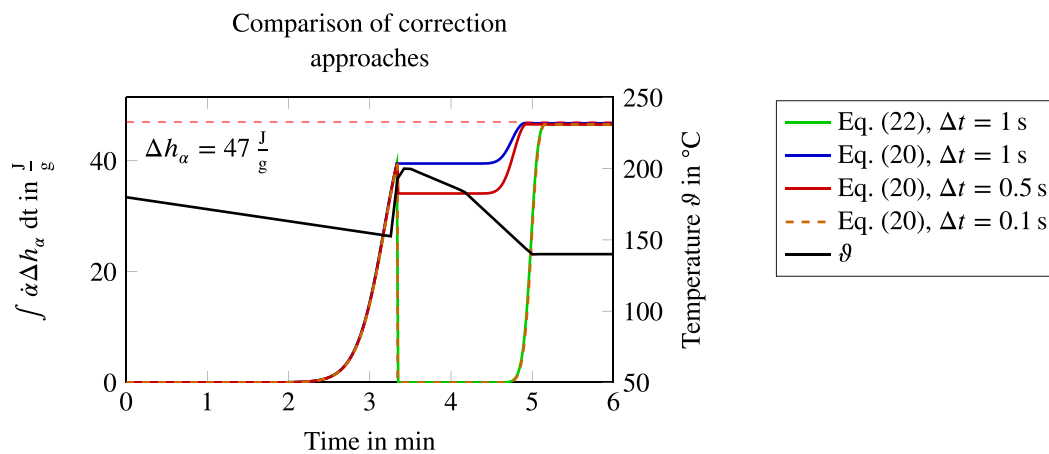


FIGURE 7 Comparison of the approach from Equations (20) and (22) for different time step sizes Δt . For a not sufficiently small time step size Equation (20) might skip over the temperature range where $\frac{d\omega}{d\theta} < 0$ and the degree of crystallization would not be reduced sufficiently. As in Figure 6 at a fully crystallized state, the accumulated released heat matches the total energy released during the exo-thermal reaction, that is, Δh_α

This can lead to a premature prediction of full crystallization, as the unmodified model reaches the fully crystallized state sooner than the modified model. Furthermore, if a mechanical model is used, with material parameters that depend on α , the prediction of the mechanical behavior using the unmodified model would be flawed in the time where α remains unchanged even though the material undergoes remelting. This will be the focus of a future study.

The proposed modified model (Eq. (11)) allows for negative rates of α and reduces the degree of crystallization in an energetically consistent manner. This becomes apparent considering that the energy, previously released during the exo-thermal crystallization, is now absorbed during melting, which reduces the accumulated released heat due to the negative rates of α .

In the next example presented in Figure 7 the approaches of Equations (20) and (22) are compared to highlight the dependency of Equation (20) on a sufficiently small time step size. For a sufficiently small time step size of 0.1 s, the model of Equation (20) gives the same results as the one from Equation (22). However, if the time step size exceeds a certain value, here for time step sizes of 0.5 s or 1 s, the model of Equation (20) tends to jump over the temperature range within which ω is active. This certainly depends on the temperature rate, so the higher the rate, the lower the necessary time step size. Applying a slight modification as in Equation (22) circumvents this problem and leads to a complete reduction of the degree of crystallization. Hence, the later represents the preferred approach.

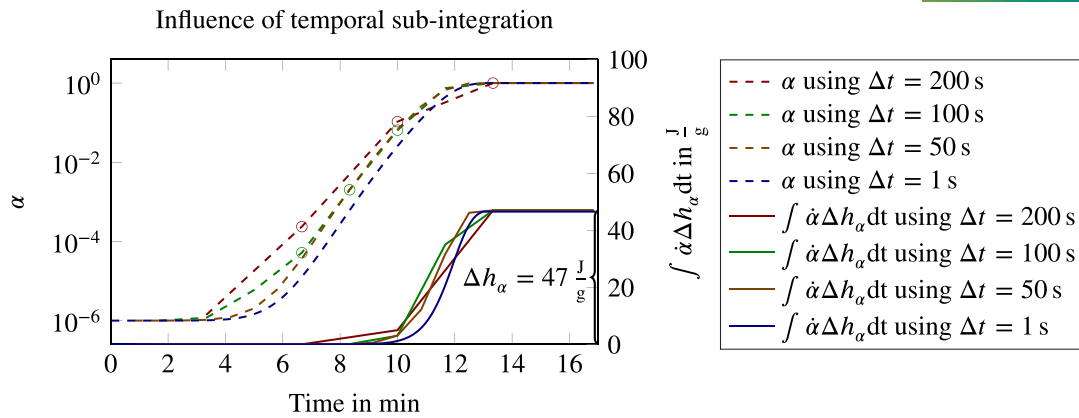


FIGURE 8 Investigating the influence of sts on the degree of crystallization α and the accumulated released heat for nonisothermal crystallization ($\vartheta_{\text{start}} = 180^\circ$, cooling rate = 2 K/min). Whenever the NR scheme on quadrature point level does not converge, the time step size on quadrature point level is reduced, that is, $\Delta t = \frac{\Delta t}{10^x}$, with $x \in \{1, \dots, n_{\text{red}}\}$ (n_{red} corresponds to the number of time step size reductions; for the present examples dividing by 10 once was sufficient). This only influences the time integration on quadrature point level. The cases where sts were used are indicated by the circular marks

4.1.2 | α evolution—Sub-time stepping

As mentioned in Section 3.3, convergence issues may occur when solving Equation (19) iteratively with a too large time step size. In these cases we allow for local sub-time stepping (sts) where the time step size on quadrature point level is divided by a certain number and the degree of crystallization is computed in a sequential manner for the subintervals. The temperature for the subintervals is thereby interpolated linearly.

In the present example the initial temperature of $\vartheta_0 = 180^\circ$ is held constant for 100 s followed by a cooling with a rate of 2 K/min. The results are depicted in Figure 8 showing the evolution of the degree of crystallization α and the accumulated released heat over time. The evolution of the degree of crystallization is shown on a logarithmic axis and the time steps where sts is required are indicated by circular marks. Whenever the NR scheme on quadrature point level does not converge, the time step size on quadrature point level is reduced, that is, $\Delta t = \frac{\Delta t}{10^x}$, with $x \in \{1, \dots, n_{\text{red}}\}$ (n_{red} corresponds to the number of time step size reductions and for the present examples dividing by 10 once was sufficient). This time step size reduction only influences the time integration on quadrature point level, while the global system is solved using the unreduced time step size. Employing sub-time stepping allows the use of even very large time step sizes while being energetically consistent. This can be observed by considering the evolution of the accumulated released heat reaching the same value for all considered time step sizes. These large time step sizes are usually only relevant during the cooling stage of the SLS process, which can take several hours in the actual process.

4.2 | Simplified quasi 1D example

In this example a simplified version of the SLS process is simulated, solving a quasi one-dimensional problem depicted in Figure 9. The intention of this example is to demonstrate the possibility and suitability of sub-time stepping due to large time step sizes in the cooling stage. In essence several elements (similar to the one-element-test but with an edge length of 100 μm) are stacked on top of each other. The boundaries are subject to homogeneous Neumann type conditions with the exception of the lower boundary (indicated via ϑ_D) and the upper boundary in build direction which is subject to radiation and convection.

During the preheating and building stages ϑ_D is set to 115° , $\vartheta^{\text{conv}} = 150^\circ$, with $H = 15 \text{ W}/(\text{m}^2\text{K})$ and the target temperature for the PID controlled radiation boundary condition is 172° . During the cooling stage, transient Dirichlet and convection boundary conditions are employed. The temperature as well as the degree of crystallization is recorded at the point MP.

The results are shown in Figures 10 and 11. In the preheating stage the initial powder material, consisting of 200 layers (or elements) each 100 μm in height, is heated up for 1 h, starting from a value of $\vartheta_0 = 35^\circ$ using the PID controlled

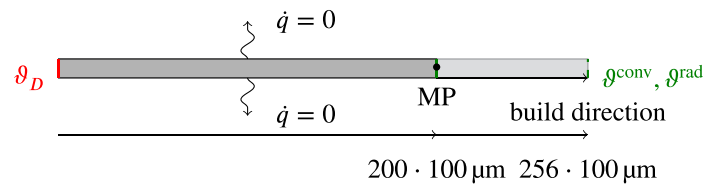


FIGURE 9 Setup for the quasi 1D example (for illustration purposes, the picture is rotated by 90°). 200 elements, as shown in the one-element-test (but with edge length $100 \mu\text{m}$), are stacked on top of each other, where each element represents a powder layer. After a preheating stage of 1 h, 56 powder layers are deposited in build direction and molten. During the preheating and build stage ϑ_D is set to 115° , $\vartheta^{conv} = 150^\circ$, with $H = 15 \text{ W}/(\text{m}^2 \text{ K})$ and the target temperature for the PID controlled radiation boundary condition is 172° . During the cooling stage, transient Dirichlet and convection boundary conditions are employed as shown in Figure 10. The temperature as well as the degree of crystallization are recorded at the point MP

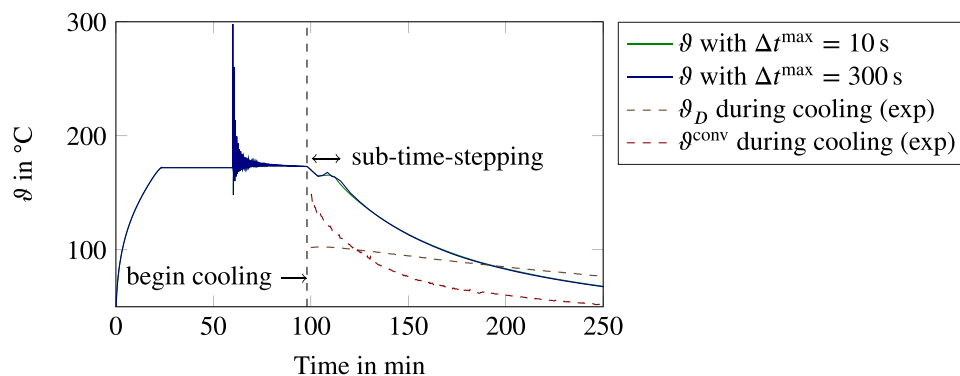


FIGURE 10 Studies for sub-time steps. Cool down started at $t \approx 98 \text{ min}$. The case with $\Delta t^{\max} = 10 \text{ s}$ during cooling needed in total 1002 time steps while the case using $\Delta t^{\max} = 300 \text{ s}$ during cooling required 40 time steps. The exo-thermal reaction at the beginning of the cooling stage leads to a temperature plateau. As indicated, sub-time stepping occurred for the case with $\Delta t^{\max} = 300 \text{ s}$

radiation boundary condition proposed in [26]. Thereafter, a total of 56 layers are deposited and molten. Eventually the process chamber is cooled down for approximately 3 h using a transient Dirichlet type boundary condition prescribing ϑ_D and convection on the upper surface in build direction. The values for the prescribed boundary conditions during the cooling stage, that is, ϑ_D and ϑ^{conv} in Figure 10, correspond to measured data and are taken from [13].

During the preheating and in the time before and after layer depositions in the building stage, the maximum time step size is set to 10 and 5 s respectively, while using the 2 stage S-DIRK inbuilt error estimator for temporal adaptivity. This ensures that the PID controller works properly and that the effects of layer deposition and successive reheating are sufficiently resolved in time. The time step sizes during melting and layer deposition are fixed at $1 \times 10^{-4} \text{ s}$ and $2.5 \times 10^{-2} \text{ s}$ respectively.

In the cooling stage, where no PID controlled radiation boundary condition is necessary, the adaptive time stepping is allowed to potentially increase the time step size even further, which might lead to sts. Therefore, three scenarios are compared: a maximum time step size during cooling of 10 s or 300 s and the case of no correction of the crystallization degree ($\omega = 1$) using 10 s as maximum time step size during cooling.

Due to the nature of the quasi one-dimensional example, the effect of the exo-thermal reaction during crystallization in the beginning of the cooling stage is clearly visible. In the region where sts occurs, the temperature rests at a plateau, while the temperature at the boundary is steadily reduced. The case where the time step size during cooling is limited to 10 s requires 1002 time steps for the cooling stage, while the approach using a maximum time step size of 300 s requires only 40 steps. The difference of the number of time steps (962) correlates with the difference in the maximum allowed time step size, that is, a factor of $\frac{300}{10} = 30$ times the needed steps for the large time step size, that is, 40, which renders 1200. This represents a significant reduction in computational cost for a comparable accuracy, which is made possible by the subtime-stepping integration scheme.

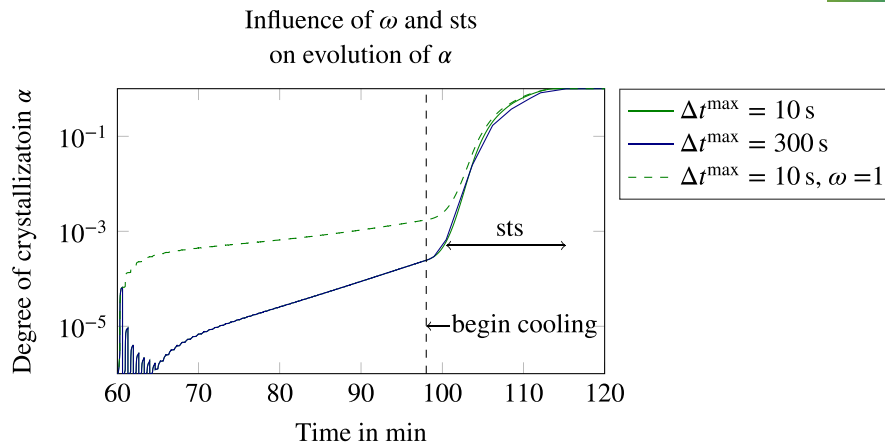


FIGURE 11 Degree of crystallization corresponding to Figure 10, including a comparison to the unmodified crystallization model of Equation (8) (indicated by $\omega = 1$)

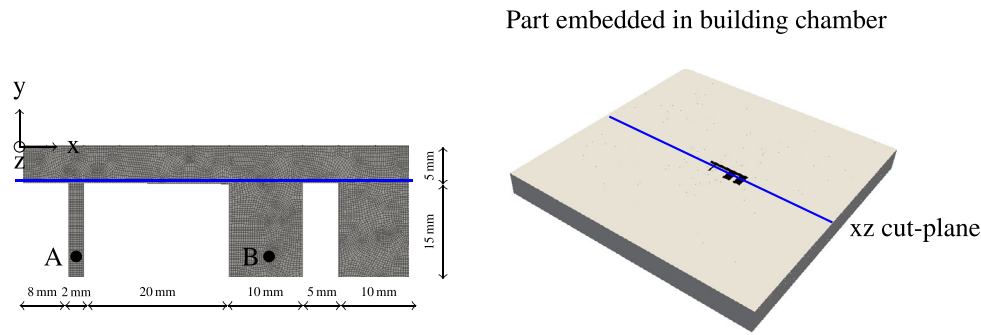


FIGURE 12 Setup for the study of feature size effects. The overall size of the part is $55 \times 20 \times 2 \text{ mm}^3$ with one thin feature of 2 mm width and two of 10 mm width. The cross sectional size of the building chamber is $350 \times 350 \text{ mm}^3$. The smallest element edge length was set to $100 \mu\text{m}$. A line length of $\tau = 700$ was used with a total scan path length of approximately 60 m for 20 layers. A meander scan strategy in x direction was applied for the melting

In Figure 11 the effects of the correction term are highlighted. While the unmodified crystallization model prohibits the reduction of α upon melting, the proposed modified ansatz behaves as expected. Besides the effect on the thermal problem, this could potentially influence mechanical predictions, since the mechanical material parameters would depend on the temperature and degree of crystallization. This will be the subject of future studies.

4.3 | Feature size dependency

In the last example the evolution of the temperature and degree of crystallization for the part presented in Figure 12 are computed. Its overall size is $55 \times 20 \times 2 \text{ mm}^3$ with one thin feature of 2 mm width and two of 10 mm width. The part is embedded in a powder bed with an initial powder bed height of 30 mm and a cross sectional size of $350 \times 350 \text{ mm}^3$. The smallest element edge-length is set to equal the layer height, that is, $100 \mu\text{m}$ and a maximum edge length of 1 mm is allowed inside the part. The mesh is coarsened away from the part and larger element edge lengths are allowed. The beam radius r_b and the hatching distance h_s are equal to $200 \mu\text{m}$. A meander scan strategy in x direction is applied for the melting. The integrated heat source uses a relative line length of $\tau = 700$. This corresponds to a scan path length of 140 mm for a beam radius of $r_b = 200 \mu\text{m}$, that is melted within one time step. After the melting of 20 layers, only fresh powder is deposited without a successive melting. This mimics for example the effect of creating distance to a possible second part that would lie on top of the first part.

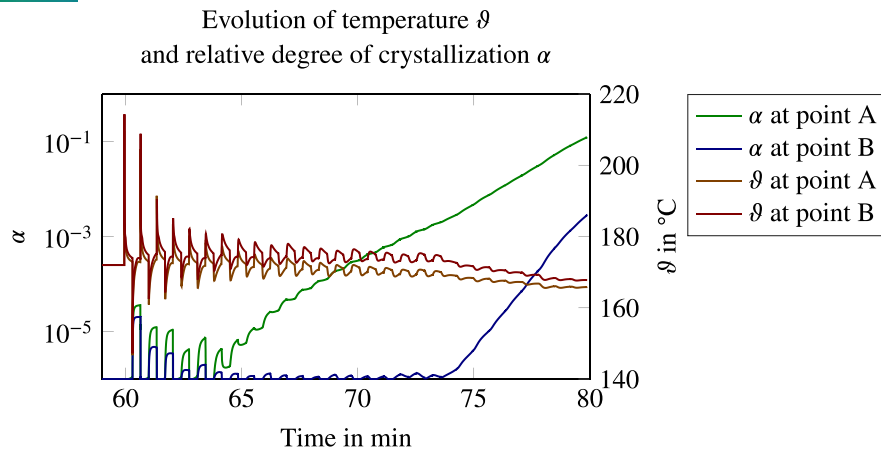


FIGURE 13 Evolution of the temperature ϑ and the relative degree of crystallization α at the two points A and B indicated in Figures 12 and 16. After 1 hour of preheating 20 layers were deposited and melted. To mimic the effect of a moving cross section or simply space between additional parts, 10 additional layers that are not being melted were deposited afterwards. Due to the spatial adaptivity slight variations can be observed for the degree of crystallization

The top surface is subject to a PID-controlled radiation boundary condition using a target temperature $\vartheta^{\text{rad}} = 172^\circ$ and convection boundary condition with $\vartheta^{\text{conv}} = 150^\circ$ and a heat convection coefficient of $\mathcal{H} = 15 \text{ W}/(\text{m}^2 \text{ K})$ (see [25,26] for details of the adaptivity procedure and the PID controller). The remaining boundary surfaces are subject to Dirichlet type boundary conditions with $\vartheta^D = 115^\circ$. The initial temperature is set to $\vartheta_0 = 35^\circ$ and the preheating time to 1 h. The process parameters are the laser power of $P_b = 16 \text{ W}$ and beam velocity $v_b = 2 \text{ m/s}$. Temperatures and degree of crystallization are recorded at the height of the initial layer at the two points A and B indicated in Figure 12. The blue line indicates a cut-plane, that will be used to show the internal distributions of temperature and degree of crystallization. It should be mentioned that the field of the degree of crystallization is obtained in a postprocessing step from the values from the quadrature points using a discontinuous Galerkin approach. Furthermore, this example required a total number of 3655 time steps, with an average number of 4 Newton iterations and an average number of degrees of freedom of ≈ 170.000 .

4.3.1 | Temperature evolution

Figure 13 shows the evolution of the temperature and degree of crystallization at the points A and B. To highlight the actual process, the preheating time of 1 h is cropped out and the plot starts at 60 min. It can be observed that the temperature at point B is higher than the one at point A, resulting in a lower degree of crystallization. These differences are mainly due to different distances of the points A and B to the surrounding powder and the neighboring feature close to point B. For both points the effects of the correction of the crystallization, that is, the repeated reduction of the degree of crystallization upon melting of the upper layer using Equation (11), is apparent.

Figure 14 gives a view into the building chamber during the three phases (i) melting, (ii) deposition of a fresh powder layer and (iii) reheating to a desired build temperature (from left to right). The molten material is shown fully, while the building chamber is cut in half in the xz -plane (see the blue line in Figure 12). This allows to not only visualize the temperature of the part itself but also the surrounding powder bed. It can be observed, that the deposition of the fresh powder significantly reduces the temperature in the upper layers. This can also be seen in Figure 13, when after the sharp increase in temperature during melting, a nearly similarly large temperature drop follows during the powder deposition.

In Figure 15 the printed part is shown from the bottom at two distinct time instances. The first ($t \approx 70 \text{ min}$) is just before the deposition of a new powder layer and the second ($t \approx 80 \text{ min}$) shortly after the deposition of the 10th isolation powder layer, that is, a layer that is not being melted.

During the melting stage, shown in the left picture, the temperature is at a higher value than during the deposition of isolation layers (right). This affects the crystallization as can be seen in Figure 13. With the beginning of the deposition of isolation powder layers ($t \approx 74 \text{ min}$) the degree of crystallization grows faster than during the melting of powder layers. Furthermore, the dependence of the heat dissipation on the geometry of the part becomes more apparent in Figure 15,

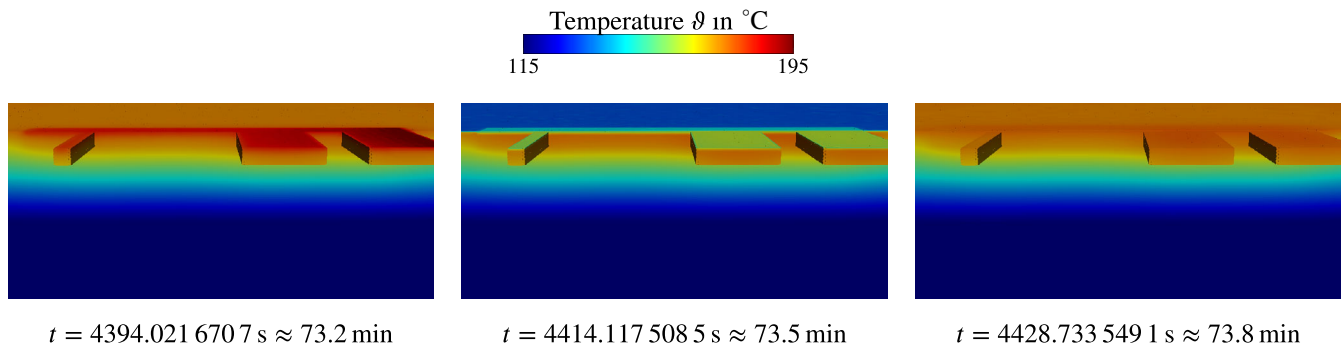


FIGURE 14 View into the building chamber at various time instances. The powder bed is cut in the xz plane (as indicated in Figure 12) and the molten material is fully shown. From left to right the first picture depicts the temperature at the end of the melting of the current cross section. The second picture illustrates the temperature immediately after the deposition of a fresh powder layer. The third picture shows the temperature after the reheating stage before the next layer is molten

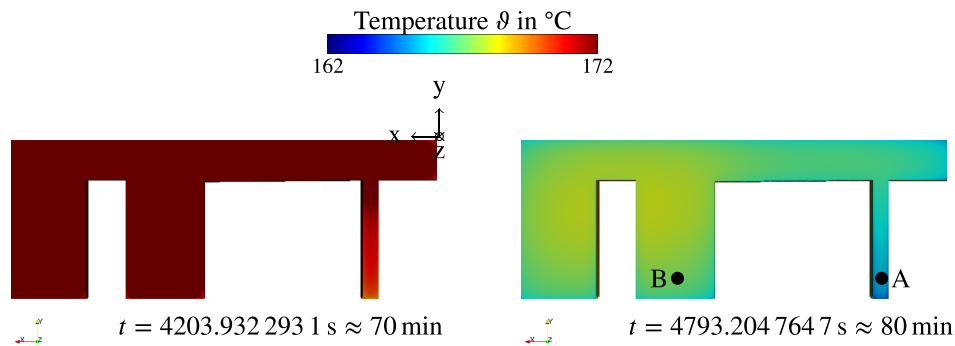


FIGURE 15 View from the bottom of the part at two time instances showing the temperature distribution. The first time instance ($t \approx 70\,\text{min}$) is during the melting stage. The second is during the deposition of layers that are not molten, so-called isolation layers (see also Figure 13)

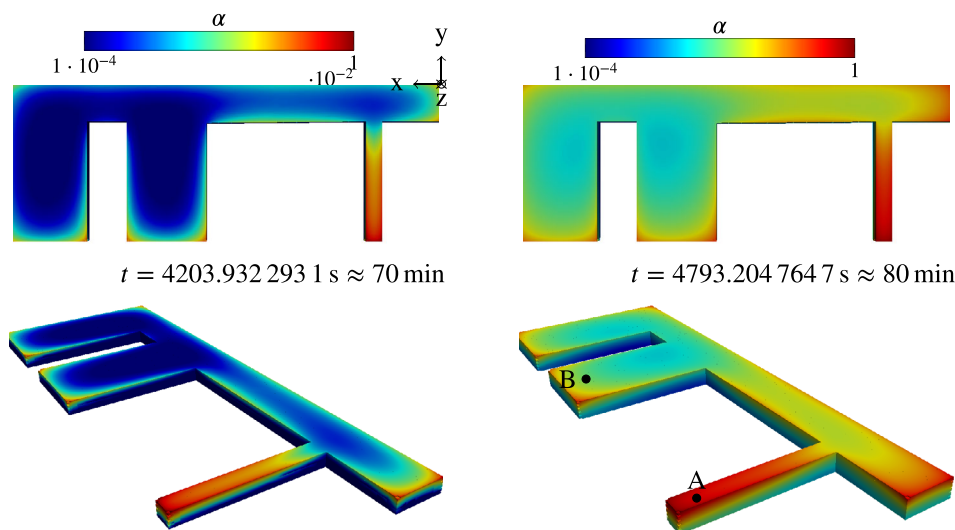


FIGURE 16 View from the bottom of the part at two time instances showing the distribution of the relative degree of crystallization α . The first time instance ($t \approx 70\,\text{min}$) is during the melting stage. The second is during the deposition of layers that are not melted, so-called isolation layers (see also Figure 13)

especially for the picture on the right. The thicker geometric features (including point B) cool down slower than the thinner one (including point A). This can be also seen in Figure 13.

4.3.2 | Crystallization degree evolution

In addition to the temperature fields the relative degree of crystallization of the part at the two aforementioned time instances, that is, $t \approx 70$ min and $t \approx 80$ min, is presented in Figure 16 (observe the different orders of magnitudes). Again, the part is shown from the bottom. The crystallization fields, similarly as the corresponding temperature fields, show a size dependency. This can be seen by comparing the development of the thin feature (2 mm width) and the two larger features (10 mm width). The larger areas show a relatively low degree of crystallization α , while smaller areas and the boundary of the part show higher values of α . The effect of not continuing to melt the powder after 20 layers becomes apparent considering the increase of the overall crystallization activity (see also Figure 13).

In the presented example the dependency of the temperature field and correspondingly the degree of crystallization on the geometrical dimensions is shown. Features that are rather thin show a lower temperature level than larger features. In the next step it would be necessary to investigate compensation strategies to render a more homogeneous crystallization behavior. One possibility would be to increase the energy input in areas with a smaller surface to volume ratio. This will be part of future work.

5 | CONCLUSION

In this work we investigated the crystallization behavior of a thermoplastic material (Polyamide 12) during selective laser sintering on a macroscopic scale using a finite element tool. To account for the typical temperature fields arising in selective laser sintering, the classic Nakamura model was extended to allow for a reduction of the degree of crystallization upon melting in an energetically consistent manner.

It has been shown that local sub-time stepping on quadrature point level represents a sufficiently accurate approximation for the temporal integration of the degree of crystallization, while being more efficient than a global constraint on the time step size. Furthermore, the influence of geometrical dimensions on the temperature- and crystallization fields has been presented. Finer features cool significantly faster than larger features which immediately affects the evolution of the degree of crystallization.

In future work, we will extend the model by the description of the mechanical material behavior and investigate possible compensation strategies to cope with the size dependency, including further experimental validation.

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REFERENCES

- [1] A. Amado, K. Wegener, M. Schmid, and G. Levy, *Characterization and modeling of non-isothermal crystallization of polyamide 12 and co-polypropylene during the SLS process*, Proceedings of the 5th International Polymers and Moulds Innovations Conference, Ghent, 2012.
- [2] D. Arndt, W. Bangerth, T. C. Clevenger, D. Davydov, M. Fehling, D. Garcia-Sanchez, G. Harper, T. Heister, L. Heltai, M. Kronbichler, and R. M. Kynch, The deal.II library, version 9.1, J. Numer. Math. **27** (2019), 203–213.
- [3] C. Balemans, S. F. Looijmans, G. Grosso, M. A. Hulsen, and P. D. Anderson, Numerical analysis of the crystallization kinetics in SLS, Addit. Manuf. **243** (2020), 101126.
- [4] S. Bhandari and R. A. Lopez-Anido, Discrete-event simulation thermal model for extrusion-based additive manufacturing of PLA and ABS, Materials **13** (2020), 4985.
- [5] R. Brighenti, M. P. Cosma, L. Marsavina, A. Spagnoli, and M. Terzano, Laser-based additively manufactured polymers: A review on processes and mechanical models, J. Mater. Sci. **56** (2021), 961–998.
- [6] A. Das, C. A. Chatham, J. J. Fallon, C. E. Zawaski, E. L. Gilmer, C. B. Williams, and M. J. Bortner, Current understanding and challenges in high temperature additive manufacturing of engineering thermoplastic polymers, Addit. Manuf. **34** (2020), 101218.
- [7] D. Drummer, S. Greiner, M. Zhao, and K. Wudy, A novel approach for understanding laser sintering of polymers, Addit. Manuf. **27** (2019), 379–388.

- [8] P. Ellsiepen and S. Hartmann, Remarks on the interpretation of current non-linear finite element analyses as differential-algebraic equations, *Int. J. Numer. Methods Eng.* **51**(6) (2001), 679–707.
- [9] S. Felder, H. Holthausen, S. Hesseler, F. Pohlkemper, T. Gries, J. W. Simon, and S. Reese, Incorporating crystallinity distributions into a thermo-mechanically coupled constitutive model for semi-crystalline polymers, *Int. J. Plast.* **135** (2020), 102751.
- [10] M. Ganci, W. Zhu, G. Buffa, L. Fratini, S. Bo, and C. Yan, A macroscale FEM-based approach for selective laser sintering of thermoplastics, *Int. J. Adv. Manuf. Technol.* **91** (2017), 3169–3180.
- [11] A. Gosset, D. Barreiro-Villaverde, J. C. Becerra Permy, M. Lema, A. Ares-Pernas, and M. J. Abad López, Experimental and numerical investigation of the extrusion and deposition process of a poly(lactic acid) strand with fused deposition modeling, *Polymers* **12** (2020), 2885.
- [12] J. Irwin and P. Michaleris, A line heat input model for additive manufacturing, *J. Manuf. Sci. Eng.* **138** (2016), 111004.
- [13] A. Jaksch, Selektives laserstrahlschmelzen von Kunststoffen: Ganzheitliche erforschung des orts aufgelösten alterungsverhaltens von polyamid 12 pulver, Master's Thesis, Institute of Polymer Technology (LKT), Friedrich-Alexander-Universität Erlangen-Nürnberg, 2016.
- [14] A. Kelly, A. P. Stebner, and K. Bhattacharya, A micromechanics-inspired constitutive model for shape-memory alloys that accounts for initiation and saturation of phase transformation, *J. Mech. Phys. Solids* **97** (2016), 197–224.
- [15] K. Krabbenhoft, L. Damkilde, and M. Nazem, An implicit mixed enthalpy-temperature method for phase-change problems, *Heat Mass Transf.* **43** (2006), 233–241.
- [16] A. Levy, *A novel physics node for Nakamura crystallization kinetics*, Proceedings of the 2016 COMSOL Conference, Munich, 2016.
- [17] A. Levy, Robust numerical resolution of Nakamura crystallization kinetics, *Int. J. Theor. Appl. Math.* **3** (2017), 143.
- [18] A. Lion and M. Jöhrlitz, A thermodynamic approach to model the caloric properties of semicrystalline polymers, *Contin. Mech. Thermodyn.* **28** (2016), 799–819.
- [19] F. M. Mwanja, M. Maringa, and J. G. van der Walt, A review of the techniques used to characterize laser sintering of polymeric powders for use and re-use in additive manufacturing, *Manuf. Rev.* **8** (2021), 14.
- [20] K. Nakamura, T. Watanabe, K. Katayama, and T. Amano, Some aspects of nonisothermal crystallization of polymers. I. Relationship between crystallization temperature, crystallinity, and cooling conditions, *J. Appl. Polym. Sci.* **16** (1972), 1077–1091.
- [21] R. M. Patel and J. E. Spruiell, Crystallization kinetics during polymer processing-analysis of available approaches for process modeling, *Polym. Eng. Sci.* **31** (1991), 730–738. <https://doi.org/10.1002/pen.760311008>
- [22] P. Peyre, Y. Rouchausse, D. Defauchy, and G. Régnier, Experimental and numerical analysis of the selective laser sintering (SLS) of PA12 and PEKK semi-crystalline polymers, *J. Mater. Process. Technol.* **225** (2015), 326–336.
- [23] D. Riedlbauer, M. Drexler, D. Drummer, P. Steinmann, and J. Mergheim, Modelling, simulation and experimental validation of heat transfer in selective laser melting of the polymeric material PA12, *Comput. Mater. Sci.* **93** (2014), 239–248.
- [24] F. Shen, S. Yuan, C. K. Chua, and K. Zhou, Development of process efficiency maps for selective laser sintering of polymeric composite powders: Modeling and experimental testing, *J. Mater. Process. Technol.* **254** (2018), 52–59.
- [25] D. Soldner and J. Mergheim, Thermal modelling of selective beam melting processes using heterogeneous time step sizes, *Comput. Math. Appl.* **78** (2019), 2183–2196.
- [26] D. Soldner, S. Greiner, C. Burkhardt, D. Drummer, P. Steinmann, and J. Mergheim, Numerical and experimental investigation of the isothermal assumption in selective laser sintering of pa12, *Addit. Manuf.* **37** (2021), 101676.
- [27] J. W. Stansbury and M. J. Idacavage, 3D printing with polymers: Challenges among expanding options and opportunities, *Dent. Mater.* **32** (2016), 54–64.
- [28] X. Tian, G. Peng, M. Yan, S. He, and R. Yao, Process prediction of selective laser sintering based on heat transfer analysis for polyamide composite powders, *Int. J. Heat Mass Transf.* **120** (2018), 379–386.
- [29] H. Xia, J. Lu, and G. Tryggvason, Fully resolved numerical simulations of fused deposition modeling. part II–solidification, residual stresses and modeling of the nozzle, *Rapid Prototyp. J.* **24** (2018), 973–987.
- [30] M. Zhao, K. Wudy, and D. Drummer, Crystallization kinetics of polyamide 12 during selective laser sintering, *Polymers* **10** (2018), 168.

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